

**Universität Stuttgart**  
Master of Science

# Distributed Systems

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# Goal

The goal of the course is to enable the student to understand the foundations of distributed systems, fundamental distributed algorithms, and to be able to design and develop such systems.

Build Distributed Systems that are:

- correctly functioning
- performance-oriented
- and reliable

Design principles of distributed systems and their application to the modern networked environment

- Understand fundamental distributed systems theory
- Design for Internet/Web/Pervasive environments

# Outline

## Tentative

### Basics:

- Characterization of Distributed Systems

- System Models

- ~~Networking~~

### Middleware:

- Interprocess Communication

- Distributed Objects and Remote Invocation

- ~~Distributed File Systems~~

- ~~Distributed Transactions~~

### Distributed Algorithms:

- Time Logical

- Time Physical

- Coordination

- Agreement

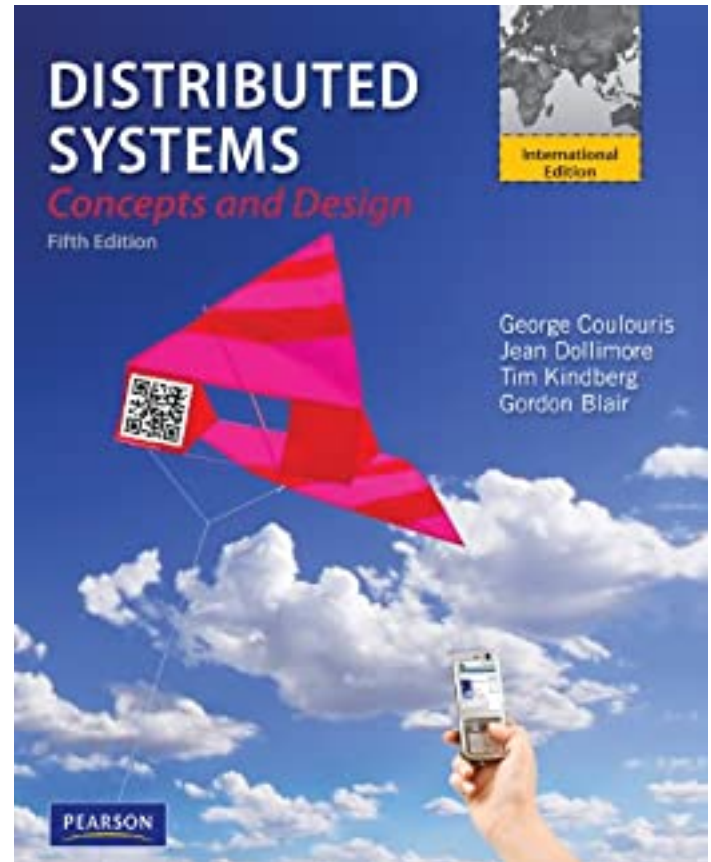
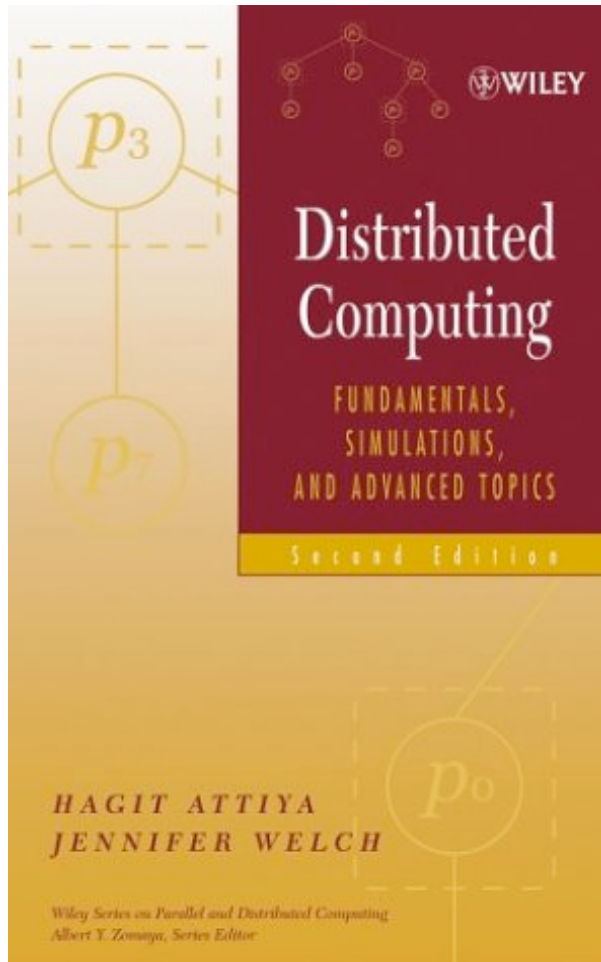
- Fault Tolerance

### System Infrastructure:

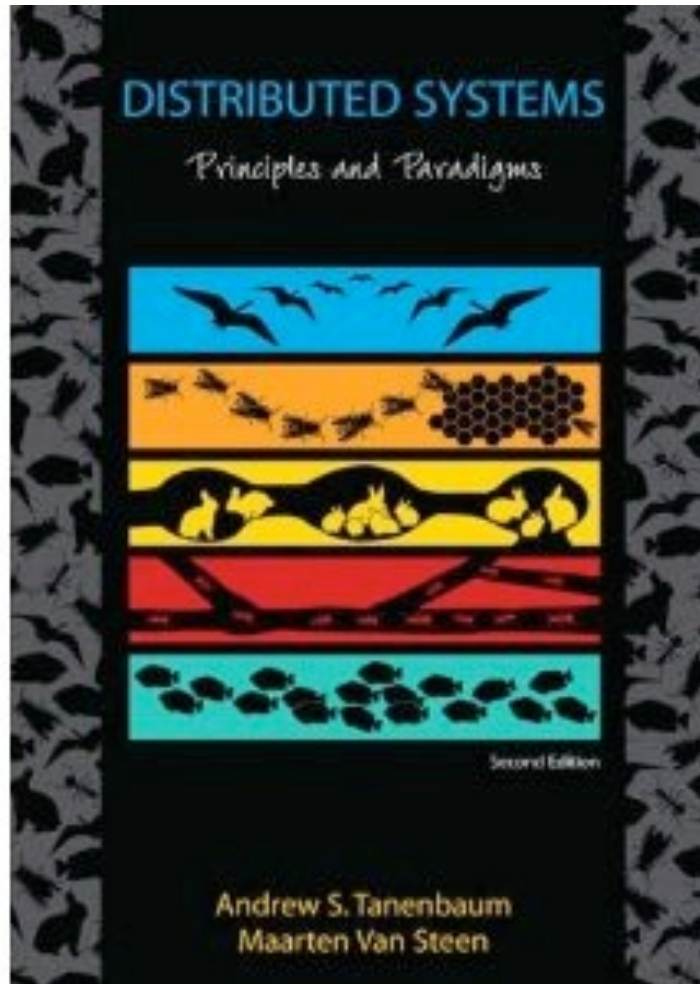
- Replication

~~Optional~~ Not mandatory for HHZ students

# Books



# A possible alternative book



# Slides are not intended for self-study

- Slides are given as a reference to help you take notes; they are not study material and do **not substitute** books, lecture notes, or other resources.



# Important Dates

|                            |                                             |
|----------------------------|---------------------------------------------|
| <b>07.05.26</b>            | Deadline to submit Preliminary Project Form |
| <b>11.05.26</b>            | Lectures and practicals                     |
| <b>12.05.26</b>            | Lectures and practicals                     |
| <b>25.05.26</b>            | Optional project description update         |
| <b>01.06.26</b>            | Lectures and practicals                     |
| <b>02.06.26</b>            | Lectures and practicals                     |
| <b>2.7.2026 (midnight)</b> | Deadline to submit Project Report Form      |
| <b>09.07.26</b>            | Tryout Exam + Project Demos                 |
| <b>16.07.26</b>            | Written Exam                                |

# Evaluation

HHZ

- Project report and demo (pass/no pass)
- Written exam (marked)



# Project Requirements

HHZ

- The Architectural Description is illustrated
- At least two distinct physical machines are used for the Demo
- Project requirements:
  - Any distributed application must implement:
    1. Dynamic discovery of hosts
    2. Crash fault tolerance
    3. Election (ring left/right communication, no list of servers!)

# Project

## How to work

- Work in groups of three
- You are responsible for forming the groups
- All members of the group pass or don't pass
- We do not intervene in group dynamics
- In Relax, only one member of the team submits; not all members need to submit the same document

## On the use of ChatGPT and similar tools



- You can use whatever tool you want for the report and coding, ChatGPT, Grammarly, etc., but you are solely responsible for the content.
- Hallucinations, incorrect statements, and nonexistent references are **your** mistakes and can lead to failing the project.
- You must be able to explain your own code. You must be able to point at code functionality.

